Research Memo: Assessing Outlier Influence on Communication Apprehension and Public Transit Usage

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# =============================================================================
# 1. SETUP ENVIRONMENT AND LOAD DATA
# =============================================================================
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from IPython.display import Markdown
import statsmodels.api as sm

plt.rcParams['figure.autolayout'] = True
plt.rcParams['figure.dpi'] = 120

df = pd.read_csv('../../cleaned_data/merged_data.csv')

# =============================================================================
# 2. VARIABLE RECODING AND DATA CLEANING
# =============================================================================
likert_mapping = {
    "Strongly disagree": 1, "Somewhat disagree": 2,
    "Neither agree nor disagree": 3, "Somewhat agree": 4, "Strongly agree": 5
}

ca_items = ['Q1', 'Q2', 'Q3', 'Q4', 'Q5', 'Q6', 'Q13', 'Q14', 'Q15', 'Q16', 'Q17', 'Q18']
reverse_items = ['Q2', 'Q4', 'Q6', 'Q14', 'Q16', 'Q17']

for item in ca_items:
    if item in df.columns:
        df[item] = df[item].map(likert_mapping)

for item in reverse_items:
    if item in df.columns:
        df[item] = 6 - df[item]

df['ca_score'] = df[ca_items].mean(axis=1)

transit_midpoint_map = {
    'Never': 0, '0-1 days a month': 0.5, '2-4 days a month': 3,
    '4-8 days a month': 6, '8 or more days a month': 12
}
df['public_transit_days'] = df['Q26'].map(transit_midpoint_map)

df_clean = df[['ca_score', 'public_transit_days']].dropna().copy()

# OLS Regression Fit & Cook's Distance
X = sm.add_constant(df_clean['ca_score'])
y = df_clean['public_transit_days']
model_full = sm.OLS(y, X).fit()
```

```python
influence = model_full.get_influence()
cooks_d = influence.cooks_distance[0]

n = len(df_clean)
threshold = 4 / n

df_clean['cooks_d'] = cooks_d
df_clean['is_influential'] = cooks_d > threshold
n_outliers = df_clean['is_influential'].sum()

df_pruned = df_clean[~df_clean['is_influential']]
X_pruned = sm.add_constant(df_pruned['ca_score'])
y_pruned = df_pruned['public_transit_days']
model_pruned = sm.OLS(y_pruned, X_pruned).fit()
```

Research or Analytical Question

Are there any statistical outliers present in the data, and to what extent do these potential outliers exert leverage or influence on the fitted regression line between communication apprehension and public transit usage?

Answer, Response, + Summary of Results

To determine whether extreme observations distort the analysis, we evaluated bivariate regression influence using Cook's Distance (threshold $4/N = 0.0159$) across $N = 252$ complete observations.

```python
comparison_data = {
    'Model': [f'Full Model (N={n})', f'Pruned Model (N={len(df_pruned)})'],
    'Intercept': [f"{model_full.params['const']:.3f}", f"{model_pruned.params['const']:.3f}"],
    'Slope (CA Score)': [f"{model_full.params['ca_score']:.3f}", f"{model_pruned.params['ca_score']:.3f}"],
    'p-value': [f"{model_full.pvalues['ca_score']:.3f}", f"{model_pruned.pvalues['ca_score']:.3f}"],
    'R-squared': [f"{model_full.rsquared:.3f}", f"{model_pruned.rsquared:.3f}"]
}

comparison_df = pd.DataFrame(comparison_data)
display(comparison_df)
```

Table 1: Comparison of OLS regression models before and after outlier removal

	Model	Intercept	Slope (CA Score)	p-value	R-squared
0	Full Model (N=252)	6.982	-0.986	0.001	0.043
1	Pruned Model (N=245)	7.849	-1.459	0.000	0.094

```python
fig, ax = plt.subplots(figsize=(5.5, 2.2))

ax.scatter(df_clean[~df_clean['is_influential']]['ca_score'],
           df_clean[~df_clean['is_influential']]['public_transit_days'],
           color='blue', alpha=0.5, s=20, label='Normal observations')

ax.scatter(df_clean[df_clean['is_influential']]['ca_score'],
           df_clean[df_clean['is_influential']]['public_transit_days'],
           color='red', s=35, edgecolors='black', label='Outliers (Cook\'s D > 4/N)')

x_vals = np.linspace(df_clean['ca_score'].min(), df_clean['ca_score'].max(), 100)
y_full = model_full.params['const'] + model_full.params['ca_score'] * x_vals
y_pruned = model_pruned.params['const'] + model_pruned.params['ca_score'] * x_vals

ax.plot(x_vals, y_full, 'b-', linewidth=1.2, label='Full Model Line')
ax.plot(x_vals, y_pruned, 'g--', linewidth=1.2, label='Pruned Model Line')
```

```python
ax.set_xlabel('Communication Apprehension Score', fontsize=7.5)
ax.set_ylabel('Public Transit Days (Monthly)', fontsize=7.5)
ax.set_title('Regression Sensitivity: Impact of High-Influence Outliers', fontsize=8.5)
ax.legend(fontsize=6, loc='upper right')
ax.grid(True, alpha=0.2)

plt.tight_layout()
plt.show()
```

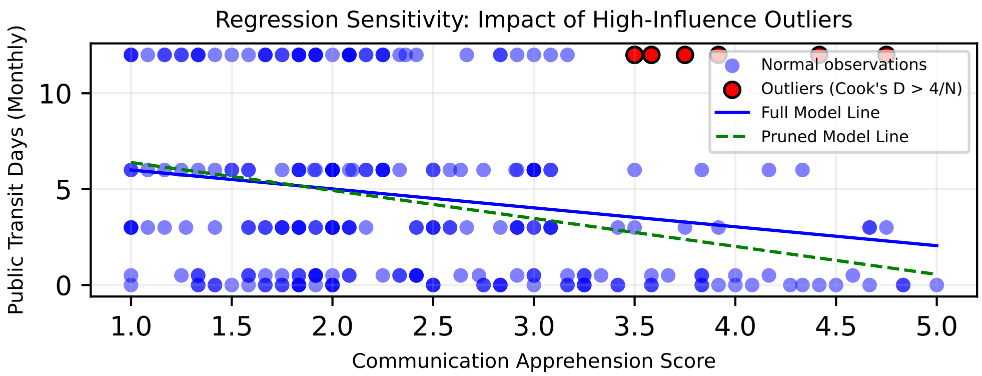

Figure 1: Scatter plot with regression lines before and after outlier removal

Summary of Results

- **Key Finding**: As shown in Table 1, we identified **7 influential observation(s)** ($4/N = 0.0159$). Removing them strengthened the negative slope from **-0.986** ($p = 0.001$) to **-1.459** ($p < 0.001$), with R^2 moving from **0.043** to **0.094**.
- **Conclusion**: As illustrated in Figure 1, removing high-influence outliers does not alter the core finding: communication apprehension has a **statistically significant negative relationship** with transit usage. Removing outliers reveals a stronger underlying negative effect.

What questions or uncertainties remain?

1. **Measurement Limitations**: Converting ordinal category choices into exact monthly days via midpoints creates discrete cluster bands and extreme boundary values. Would an Ordinal Logistic Regression (or a Binary Logistic model for regular usage) better accommodate these ordinal categories without relying on arbitrary midpoint assumptions?
2. **Omitted Variable Bias**: The identified outliers may reflect unmeasured socio-demographic factors (e.g., vehicle ownership, transit access, income) rather than true anomalies in communication apprehension.